Hand Tremor Stabilization Update

MEDesign General Body Meeting | Duke University

Project Objectives

- **Clinical Problem:** Address involuntary resting and action tremors interfering with daily living.
- **Primary Goal:** Develop a wearable device that actively suppresses tremors without blocking intentional movement.
- **Current Focus:** Delivering a benchtop prototype implementing vibrational control.

2025-2026

Electrical Context

Real-Time Tremor Dynamics

Essential Tremor: Involuntary oscillatory motion

Typical Frequency: ~4–12 Hz

Device Requirements

- Capture motion accurately
- Process signals in real time
- Deliver corrective output

Key Challenge: Low-frequency motion + noise separation

ELECTRICAL



Mechanical Context

Electrical Component Housing

Wrist Attachment Device

Motor-Nerve Interaction

Device Requirements

- Not bulky
- Actively resists tremors once detected
- Comfortable

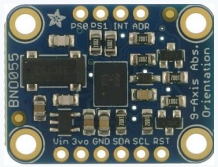
Main goal: Integrate directly with electrical team’s design function

MECHANICAL

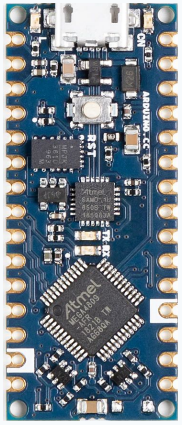
Proposed Hardware

Solution Architecture

IMU Sensor (BNO-055)
6-axis accelerometer + gyroscope
Detects tremor amplitude + frequency



Microcontroller
Arduino Nano
Handles real-time processing



Motor Driver (DRV-2605L)
Converts control signals into hardware actuation and controls signal intensity



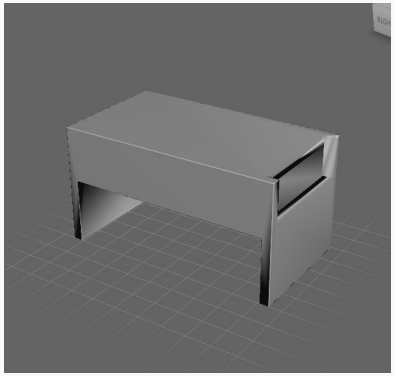


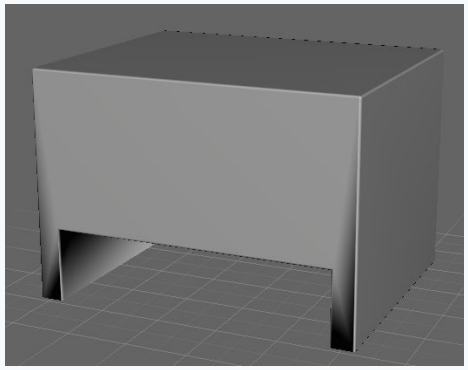
Battery System (Eventually)
Portable power supply for wearable operation - Rechargeable Li-ion battery

Vibration Motors
Haptic feedback for active tremor mitigation

SYSTEM

Hardware cont.
(Mech Components)





3D Printed Casing

Velcro Attachment

Wrist Brace

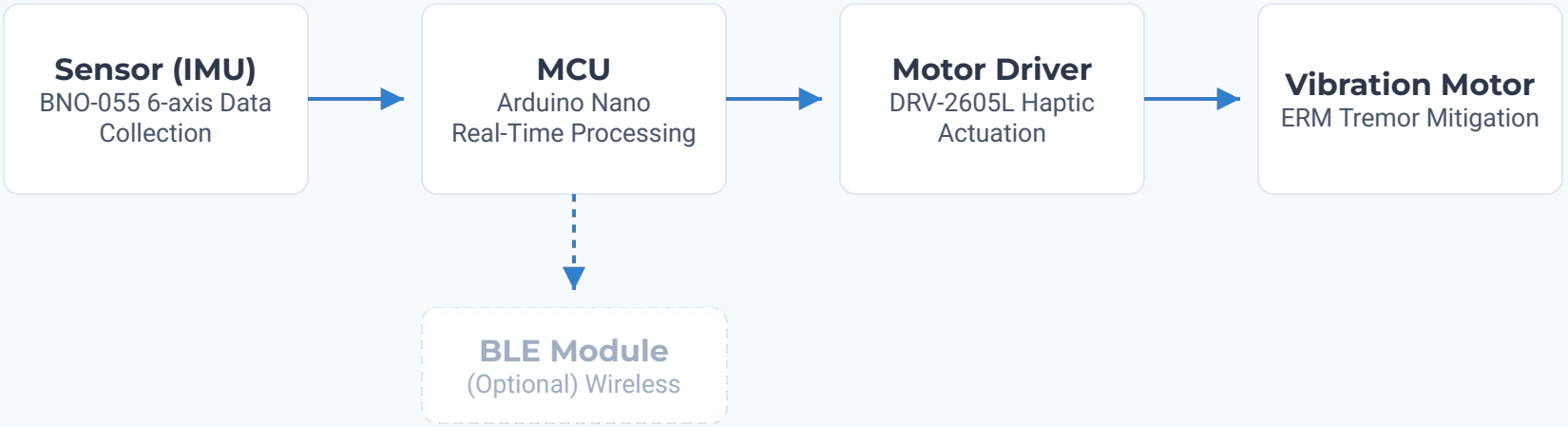




SYSTEM

System Block Diagram

Functional Signal Flow Architecture



SYSTEM ARCH

Detection Algorithm

Real-Time Processing Pipeline

Sensor Sampling 01

- IMU collects gyro data (x, y, z)
- Fixed rate (~100 Hz)

Preprocessing 02

- Offset calibration
- Digital noise filtering

Tremor Detection (FFT) 03

- Frequency domain conversion
- Target: 4–12 Hz range

Control Logic 04

- Identify strongest axis
- Calculate response timing

Haptic Output 05

- Anti-phase vibration
- Scale intensity with strength

Continuous Loop 06

- Real-time execution
- Dynamic response updates



SCRIPT LOGIC

Progress: Integration

Architecture & Connectivity Milestones

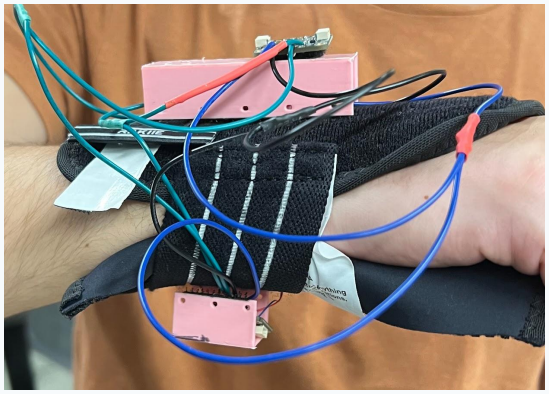
Hardware Interfaces

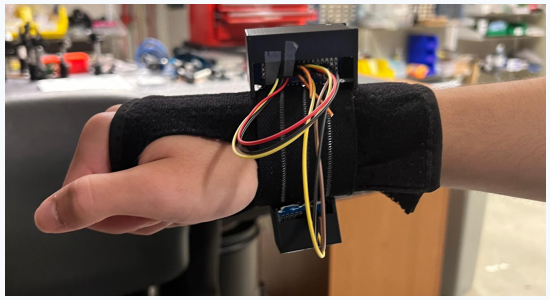
- First Working prototype (Top Right)
- Improved Electrical Harness (Bottom Right)
- Final Prototype

Team Development

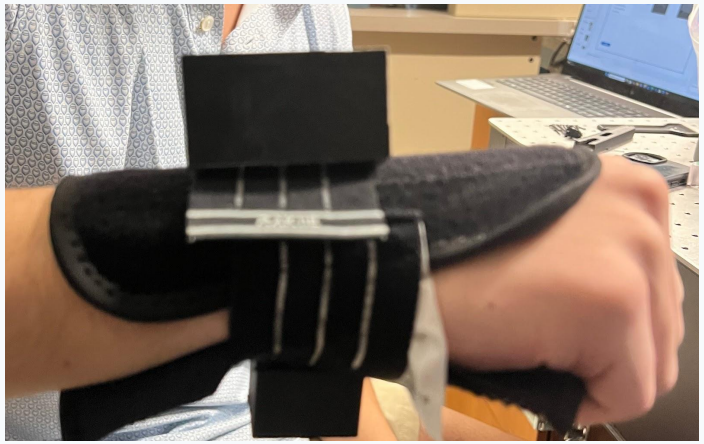
- Arduino + embedded basics training
- Detection Algorithm and Lab Skills

PROGRESS









Long-Term Goals

Strategic Development Roadmap

Performance & Connectivity

- Improve tremor suppression performance
- Add BLE + app interface
- Clinical validation potential

Optimization Targets

- Size: Reduced form factor
- Comfort: Ergonomic refinements
- Battery Life: Extended wearable operation



STRATEGY

Next Steps

Prototype & Validation Roadmap

Electrical

- PCB
- Validate IMU signal accuracy

Implementation

- Optimize Filtering and reduce latency
- Tremor detection algorithm rework

Testing & Iteration

- Test vibration response effectiveness
- Iterate: Hardware layout & Control logic

Mechanical

- Build new casing for PCB
- Create permanent attachment methods
- Create a sleeve design to test function of solely the vibrational motor
- Create a visually appealing design

EXECUTION